

Simón Madrid Alvarez

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Professional profile

Senior Data Scientist with a background in Astronomy and an M.Sc. in Analytics, specializing in fraud detection, credit risk, and early warning systems. Proven expertise in leading the end-to-end Machine Learning model lifecycle, from data architecture to production deployment. Adept at solving complex problems with limited data using weak supervision techniques, geospatial analysis, and natural language processing to mitigate financial risks and optimize business processes.

Technical skills

Languages & Libraries: Python (Pandas, NumPy, Scikit-learn, CatBoost, PyTorch, Dask, Snorkel), Advanced SQL, Bash.

Architecture & Deployment: AWS, GCP, Azure, Airflow, Git/GitHub (CI/CD), Anaconda/uv environments.

Geospatial & Graphs: H3 Hierarchical Spatial Index, Geohash, TomTom API, Neo4j.

AI & NLP: LLMs (Gemini, Instructor), Knowledge Graphs, structured information extraction.

Visualization & BI: Power BI (DAX, dynamic dashboards), Matplotlib, Seaborn.

Experience

Galgo 2025 – Present

Sr. Data Scientist (Fraud Modeling Lead) *Remote*

- Lead the end-to-end lifecycle of analytical models, overseeing data engineering, architecture, and production deployment.
- Spearheaded the development of the "fraud-checkout" model, an early alert system designed to identify fraudulent behavior in initial stages with limited client data.
- Applied advanced methodologies for imbalanced class treatments and implemented data labeling strategies using weak supervision (Snorkel) to effectively distinguish standard credit risk from intentional fraud.

Rappi 2023 – 2025

Data Scientist *Remote*

- Developed and deployed statistical and predictive machine learning models in the fraud detection domain to identify suspicious patterns and mitigate financial risk.
- Analyzed large-scale datasets to continuously improve fraud detection systems for high-volume transactions.
- Collaborated with the Special Projects & Analytics (SP&A) team to design high-impact algorithms focused on pricing, personalization, and customer experience.

Chiper 2022 – 2023

Data Scientist *Remote*

- Engineered supervised and unsupervised algorithms, neural networks, and graph theory models for automatic routing and process optimization, enhancing logistics and delivery efficiency.

AUB Capital 2021 – 2022

Quantitative Scientist *Remote*

- Processed and cleaned large financial datasets using Python and SQL.
- Executed feature engineering pipelines to build descriptive and predictive finance models utilizing Neural Networks and Reinforcement Learning algorithms.

Notable Analytical Projects

Knowledge Extraction with NLP & Graphs | *Python, Pydantic, LLMs, Neo4j*

- Built extraction pipelines using LLMs (Gemini, Instructor) to parse complex scientific literature.
- Extracted biological entities and structured ontologies, loading the transformed data into a Neo4j Knowledge Graph.

Geospatial Fraud Intelligence | *Python, H3, TomTom API*

- Implemented spatial hierarchical indexing (Uber's H3) to calculate geographic densities and identify location-based fraud patterns.

BI Automation & Automated Reporting | *Power BI, DAX, Python*

- Developed automated ETL workflows feeding dynamic Power BI dashboards to track financial KPIs.
- Created Python scripts integrated with SMTP to automate alert distributions and reporting workflows.

Education

Universidad EAFIT	Medellín, Colombia
<i>M.Sc. in Data Science and Analytics</i>	<i>2022 – 2023</i>
Universidad de Antioquia	Medellín, Colombia
<i>B.S. in Astronomy</i>	<i>2014 – 2021</i>

Publications

Search for Galactic Substructures in GAIA Data (DR2) | *Computational Physics and Astrophysics Group (FACom), Universidad de Antioquia (2019 – 2020).*